Second-Order Dissipative EFT and Frequency-Dependent Corrections to Analog Hawking Radiation in BEC Sonic Black Holes

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[1] *Theoretical Physics*

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We extend the Schwinger–Keldysh effective field theory analysis of dissipative corrections to Hawking radiation in Bose–Einstein condensate sonic black holes to second order in the derivative expansion. The first-order analysis (Paper I) established a constant dissipative correction $\delta_{\rm diss} = O(\Gamma_H/\kappa)$ parameterized by two transport coefficients (γ_1, γ_2). Here we show that at second order, exactly **two** new transport coefficients $(\gamma_{2,1}, \gamma_{2,2})$ appear, both requiring broken spatial parity (background flow). These generate frequency-dependent corrections $\delta^{(2)}(\omega) \propto (\omega/\Lambda)^3$ to the Hawking spectrum, producing a non-trivial spectral shape beyond the constant shift. A systematic monomial enumeration, verified in Lean 4 and stress-tested by the Aristotle automated theorem prover, confirms the counting formula $\text{count}(N) = \lfloor (N+1)/2 \rfloor + 1$ at each EFT order. The positivity constraint from the Schwinger–Keldysh axioms forces $\gamma_{2,1} + \gamma_{2,2} = 0$ at this truncation, reducing the independent second-order parameters to one. WKB mode analysis with dissipative turning-point shift provides the connection formula for the modified Bogoliubov coefficients, yielding the full effective temperature

$$T_{\rm eff}(\omega) = T_H \left(1 + \delta_{\rm disp} + \delta_{\rm diss} + \delta^{(2)}(\omega) + \delta_{\rm cross}\right),$$

where $\delta^{(2)}(\omega)$ is the new frequency-dependent term absent at first order.

I. INTRODUCTION

In Paper I [1], we computed the first dissipative corrections to analog Hawking radiation in BEC sonic black holes using the Schwinger–Keldysh (SK) effective field theory framework. The main result was a constant shift $\delta_{\rm diss} = O(\Gamma_H/\kappa)$ to the Hawking temperature, parameterized by two transport coefficients (γ_1, γ_2) that characterize leading-order dissipation in the superfluid.

However, the first-order analysis leaves several questions open:

1. How many new transport coefficients appear at each subsequent EFT order?
2. Do higher-order corrections introduce *frequency dependence* in the Hawking spectrum, or does the correction remain a simple temperature shift?
3. What role does spatial parity breaking (from the background flow) play in the dissipative structure?
4. How does the positivity axiom constrain the second-order parameters?

This paper addresses all four questions. The key results are:

- At EFT order N, the number of independent dissipative transport coefficients is $\lfloor (N+1)/2 \rfloor + 1$, giving 2 at first order and 2 new at second order (4 cumulative).
- The two new second-order coefficients *both require broken spatial parity*. In a homogeneous system with preserved $x \to -x$ symmetry, there are zero new coefficients at second order.
- The frequency-dependent correction $\delta^{(2)}(\omega) \propto \omega^3$ generates a spectral distortion that is in principle distinguishable from a simple temperature shift.
- The SK positivity axiom forces the constraint $\gamma_{2,1} + \gamma_{2,2} = 0$, reducing the effective number of new parameters to one.

All counting results and the algebraic KMS structure are formalized in Lean 4 and stress-tested by the Aristotle automated theorem prover.

II. SECOND-ORDER SK-EFT: MONOMIAL ENUMERATION

A. Derivative Expansion and Monomial Classification

At EFT order N, the dissipative SK action receives contributions from monomials of the form $\psi_a \cdot \partial_t^m \partial_x^n \psi_r$ (real sector) and $(\partial^\alpha \psi_a)(\partial^\beta \psi_a)$ (imaginary sector) at derivative level $L = N + 1$.

The three SK axioms constrain these monomials:

Normalization: Each monomial must contain at least one ψ_a factor. This eliminates pure ψ_r terms.

KMS symmetry (time reversal): The CGL dynamical KMS transformation includes $t \to -t$, which requires the number of time derivatives m to be even.

Fluctuation-dissipation relation: The CGL dynamical KMS condition pairs each noise coefficient with an odd-in-ω (dissipative) retarded coefficient via the master formula $K_N(\omega, k) = -i[K_R(\omega) - K_R(-\omega)]/(\beta_0 \omega)$. Even-in-$\omega$ (conservative) retarded terms are unconstrained by the FDR. The noise sector introduces no new free parameters.

After imposing these constraints, the surviving real monomials at level $L = N+1$ are indexed by pairs (m, n) with $m + n = N + 1$ and m even. The number of such pairs is

$$\text{count}(N) = \left\lfloor \frac{N+1}{2} \right\rfloor + 1. \tag{1}$$

B. First Order ($N = 1$, Level $L = 2$)

The surviving pairs are $(m, n) \in \{(0, 2), (2, 0)\}$, giving 2 transport coefficients:

$$\gamma_1 : \quad \psi_a \cdot \partial_x^2 \psi_r \quad (m = 0, n = 2) \tag{2}$$

$$\gamma_2 : \quad \psi_a \cdot \partial_t^2 \psi_r \quad (m = 2, n = 0) \tag{3}$$

This matches the result of Paper I.

C. Second Order ($N = 2$, Level $L = 3$)

The surviving pairs are $(m, n) \in \{(0, 3), (2, 1)\}$, giving 2 new transport coefficients:

$$\gamma_{2,1} : \quad \psi_a \cdot \partial_x^3 \psi_r \quad (m = 0, n = 3) \tag{4}$$

$$\gamma_{2,2} : \quad \psi_a \cdot \partial_t^2 \partial_x \psi_r \quad (m = 2, n = 1) \tag{5}$$

Critical observation: Both new monomials have *odd* spatial derivative count n. Under spatial parity $x \to -x$, these terms change sign. Therefore, in a system preserving spatial parity, $\gamma_{2,1} = \gamma_{2,2} = 0$ and there are *no new dissipative corrections* at second order.

In a BEC with transonic background flow $v(x)$, the $x \to -x$ symmetry *is* broken by the flow profile, enabling both coefficients. This is a key physical insight: the frequency-dependent correction to the Hawking spectrum is a direct signature of the background flow breaking spatial parity.

D. Formal Verification

The counting formula (1) and the specific counts at first and second order are formalized as Lean 4 theorems:

- `firstOrder_count`: Card of filter on $\text{Icc}(0, 2) \times \text{Icc}(0, 2)$ with $m + n = 2$, m even, equals 2.
- `secondOrder_count`: Card of analogous filter with $m + n = 3$ equals 2.
- `secondOrder_count_with_parity`: With the additional constraint n even (spatial parity), the count is 0.

All four counting lemmas have been formally verified by the Aristotle automated theorem prover (run `d61290fd`):

- `transport_coefficient_count`: General formula for all $N \geq 1$, proved via bijection with `Finset.range`.
- `firstOrder_count`: Verified by `native_decide` (finite decidable).
- `secondOrder_count`: Verified by `native_decide`.
- `secondOrder_count_with_parity`: Verified by `native_decide`; confirms zero coefficients survive spatial parity at second order.

These transport-coefficient counts are formally machine-verified in Lean 4.

III. FULL ALGEBRAIC KMS AT SECOND ORDER

A. The 14-Parameter General Action

The most general SK action through second derivative order has 14 independent coefficients: 9 from first order (Paper I) and 5 new at level 3 (4 real + 1 imaginary cross-term).

The complete algebraic KMS condition imposes 10 constraints:

1. Level ≤ 1 real monomials killed: $r_3 = r_4 = r_5 = r_6 = 0$ (4 constraints)
2. First-order FDR: $i_1 \cdot \beta = -r_2$, $i_2 \cdot \beta = r_1 + r_2$, $i_3 = 0$ (3 constraints)
3. Level 3 time-reversal: $s_2 = 0$, $s_4 = 0$ (2 constraints)
4. Second-order FDR: $j_{tx} \cdot \beta = s_1 + s_3$ (1 constraint)

After imposing all 10 constraints, $14 - 10 = 4$ free parameters remain, corresponding to $(\gamma_1, \gamma_2, \gamma_{2,1}, \gamma_{2,2})$.

B. Strong Uniqueness Theorem

The analog of Paper I's first-order uniqueness result:

Theorem (`fullSecondOrder_uniqueness`). *Any general SK action (parameterized by 14 real coefficients) satisfying positivity and the full algebraic KMS condition is determined by exactly 4 transport coefficients* $(\gamma_1, \gamma_2, \gamma_{2,1}, \gamma_{2,2})$ *with* $\gamma_1, \gamma_2 \geq 0$.

Aristotle (run `c4d73ca8`) proved this theorem in full. The proof constructs `CombinedDissipativeCoeffs` with $\gamma_1 = -c_{r2}$, $\gamma_2 = c_{r1} + c_{r2}$, $\gamma_{2,1} = c_{s1}$, $\gamma_{2,2} = c_{s3}$. Non-negativity of γ_1 and γ_2 follows from positivity at specific field configurations combined with the FDR relations. Lagrangian equality is verified by substituting the KMS constraints and applying `field_simp/ring`. This confirms that the second-order FDR relation $j_{tx} \cdot \beta = s_1 + s_3$ is correct — Aristotle found no counterexample.

This theorem serves as a stress test of the KMS condition. In Paper I, Aristotle discovered that the original KMS formulation was too weak (counterexample $c = (0, \ldots, 0, 1, 0)$). The same could happen here: if Aristotle finds a counterexample to `fullSecondOrder_uniqueness`, it would reveal that the second-order FDR relation $j_{tx} \cdot \beta = s_1 + s_3$ is incorrect and needs revision.

C. Positivity Constraint

The imaginary part of the combined action through second order takes the quadratic form

$$\mathrm{Im}\,\mathcal{L} = \frac{\gamma_1}{\beta}\psi_a^2 + \frac{\gamma_2}{\beta}(\partial_t\psi_a)^2 + \frac{\gamma_{2,1}+\gamma_{2,2}}{\beta}(\partial_t\psi_a)(\partial_x\psi_a). \tag{6}$$

For this to be non-negative definite for all field configurations, the associated matrix

$$M = \frac{1}{\beta}\begin{pmatrix} \gamma_1 & 0 & 0 \\ 0 & \gamma_2 & \frac{1}{2}(\gamma_{2,1}+\gamma_{2,2}) \\ 0 & \frac{1}{2}(\gamma_{2,1}+\gamma_{2,2}) & 0 \end{pmatrix} \tag{7}$$

must be positive semidefinite. With the $(3,3)$ entry equal to zero, the only way for the 2×2 lower-right submatrix to be PSD is if the off-diagonal vanishes:

$$\gamma_{2,1} + \gamma_{2,2} = 0. \tag{8}$$

This is a *strong* constraint: at this truncation order, positivity forces $\gamma_{2,2} = -\gamma_{2,1}$, reducing the independent second-order parameters to one.

Physical interpretation: The vanishing of $\gamma_{2,1}+\gamma_{2,2}$ means the second-order noise cross-term $(\partial_t\psi_a)(\partial_x\psi_a)$ has zero coefficient. This suggests that at second order, the noise structure remains diagonal in the derivative basis — a non-trivial consistency check from the positivity axiom.

Aristotle (run `c4d73ca8`) proved this constraint by contradiction: if $\gamma_{2,1} + \gamma_{2,2} \neq 0$, one constructs a field configuration with $\partial_t\psi_a = 1$ and $\partial_x\psi_a = -(\gamma_2{+}1)/(\gamma_{2,1}{+}\gamma_{2,2})$, yielding $\mathrm{Im}\,\mathcal{L} = -1/\beta < 0$, contradicting positivity. This formally establishes Eq. (8).

D. Derivation from the CGL Dynamical KMS Transformation

The algebraic FDR constraints in §III (items 2 and 4) were initially adopted as axioms, with the second-order relation $j_{tx} \cdot \beta = s_1 + s_3$ conjectured from first-order pattern matching and validated by Aristotle stress tests. We now derive these constraints from the Crossley-Glorioso-Liu (CGL) dynamical KMS transformation [5].

The CGL transformation. In the classical limit of the Schwinger-Keldysh formalism, thermal equilibrium is encoded as a $\mathbb{Z}_2$ symmetry acting on the doubled fields as

$$\tilde{\psi}_r(x) = \Theta\,\psi_r(x), \qquad \tilde{\psi}_a(x) = \Theta(\psi_a(x) + i\beta_0\,\partial_t\,\psi_r(x))\,, \tag{9}$$

where Θ is time reversal ($t \to -t$). The effective action must satisfy $I_{\mathrm{EFT}}[\psi_r, \psi_a] = I_{\mathrm{EFT}}[\tilde{\psi}_r, \tilde{\psi}_a]$.

The master FDR formula. The retarded kernel $K_R(\omega, k) = \sum_{m,n} c_{m,n}\,(-i\omega)^m\,(ik)^n$ decomposes into conservative (even-ω) and dissipative (odd-ω) parts. In Fourier space, the CGL invariance condition determines the noise kernel:

$$K_N(\omega, k) = \frac{-i}{\beta_0\,\omega}\left[K_R(\omega, k) - K_R(-\omega, k)\right] = \frac{2}{\beta_0\,\omega}\,\mathrm{Im}\big[K_R^{\mathrm{odd}}(\omega, k)\big]\,. \tag{10}$$

Even-in-ω terms cancel identically: a purely non-dissipative system correctly produces *zero* noise.

Einstein relation. At the simplest level ($L = 1$), a single dissipative monomial $-\gamma\,\psi_a\,\partial_t\psi_r$ gives $K_R = i\gamma\omega$, and the FDR yields $K_N = 2\gamma/\beta_0$, recovering the Einstein relation $\sigma = \gamma T$. This is formalized as `einstein_relation` in `CGLTransform.lean` (Aristotle run `dab8cfc1`).

General-order pairing. At each derivative level $L = 2j + 1$ (odd), every dissipative coefficient $b_{2j_t+1,2j_x}$ with $2j_t + 2j_x + 1 = L$ is paired with a noise bilinear $(\partial_t^{j_t}\partial_x^{j_x}\psi_a)^2$ via

$$i_{j_t,j_x} = -\frac{b_{2j_t+1,2j_x}}{\beta_0}. \tag{11}$$

Conservative (even-ω) coefficients at even L are unconstrained by the FDR — they derive from the free-energy functional. This is formalized as `cgl_fdr_general` and `cgl_fdr_spatial` (Aristotle run `2ca3e7e6`).

Connection to the algebraic FDR. The CGL relation Eq. (11) determines noise from *odd-ω dissipative* retarded terms, not from the even-ω conservative terms r_1, r_2 that appear in the algebraic FDR of §III. The connection proceeds through three steps:

1. *CGL FDR*: $i_1 = \gamma_{\mathrm{diss}}/\beta_0$ (noise paired with odd-ω friction)
2. *Time-reversal cancellation*: the total odd-ω retarded coefficient vanishes ($b_{\mathrm{diss}} + c_{\mathrm{conservative}} = 0$), so $b_{\mathrm{diss}} = -c_{\mathrm{conservative}}$
3. *Model identification*: for the BEC acoustic metric, $c_{\mathrm{conservative}} = -r_2$ (the conservative odd-m coefficient at level 1 is related to the spatial coefficient r_2)

Chaining: $i_1 \cdot \beta = -b_{\mathrm{diss}} = c_{\mathrm{conservative}} = -r_2$, reproducing the algebraic relation $i_1\beta = -r_2$. At second order, the same chain gives $j_{tx} \cdot \beta = s_1 + s_3$ (formalized as `cgl_implies_secondOrderKMS`, Aristotle run `2ca3e7e6`). The model identification in step 3 is specific to the BEC system; the CGL FDR (steps 1–2) is universal.

Noise count pattern. The FDR produces noise bilinears only at even EFT orders ($N = 0, 2, 4, \ldots$). At odd orders ($N = 1, 3, \ldots$), the dissipative terms generate an imaginary K_N with no real noise bilinears. The number of independent noise coefficients at even order $N = 2n$ is $n+1$, matching the transport coefficient counting formula count(N).

IV. WKB MODE ANALYSIS WITH DISSIPATIVE CORRECTIONS

A. Dissipative Dispersion Relation

Including second-order terms, the dispersion relation in natural units ($\kappa = 1$, $c_s = 1$) becomes

$$(\omega - kv)^2 = k^2(1 + D^2k^2) - i\omega\Gamma(k, \omega), \tag{12}$$

where $D = \xi\kappa/c_s$ is the dimensionless adiabaticity parameter (so the Bogoliubov quartic term carries the coefficient $D^2 = (\xi\kappa/c_s)^2$) and

$$\Gamma(k, \omega) = \gamma_1 k^2 + \gamma_2 \frac{\omega^2}{c_s^2} + \gamma_{2,1} k^3 + \gamma_{2,2}\omega^2 k \tag{13}$$

is the full damping rate through second order.

B. Turning Point Shift

The WKB turning point (where the local wavenumber diverges) shifts into the complex x-plane by

$$\delta x_{\text{imag}} = \frac{\Gamma_H}{\kappa \cdot c_s}, \tag{14}$$

where $\Gamma_H = \Gamma(k_H, \omega)$ is the damping rate evaluated at the horizon wavenumber. The connection formula across the complex turning point gives the modified Bogoliubov coefficient:

$$|\beta_\omega|^2 = \frac{1}{e^{2\pi\omega/\kappa(1+\delta_{\text{total}}(\omega))} - 1}, \tag{15}$$

where the total correction decomposes as

$$\delta_{\text{total}}(\omega) = \underbrace{\delta_{\text{disp}}}_{\text{const.}} + \underbrace{\delta_{\text{diss}}}_{\text{const.}} + \underbrace{\delta^{(2)}(\omega)}_{\text{freq.-dep.}} + \underbrace{\delta_{\text{cross}}}_{\text{small}}. \tag{16}$$

C. Frequency-Dependent Correction

The second-order contribution to the effective temperature is

$$\delta^{(2)}(\omega) = \frac{\gamma_{2,1}}{\kappa}\left(\frac{\omega}{\Lambda}\right)^3 + \frac{\gamma_{2,2}}{\kappa}\left(\frac{\omega}{\Lambda}\right)^2 \frac{\omega}{\Lambda_x}, \tag{17}$$

where $\Lambda \sim c_s/\xi$ is the EFT cutoff and $\Lambda_x \sim \kappa/c_s$ is the spatial momentum scale set by the background flow gradient.

Using the positivity constraint $\gamma_{2,2} = -\gamma_{2,1}$, this simplifies to

$$\delta^{(2)}(\omega) = \frac{\gamma_{2,1}}{\kappa}\left(\frac{\omega}{\Lambda}\right)^2 \left[\frac{\omega}{\Lambda} - \frac{\omega}{\Lambda_x}\right]. \tag{18}$$

A key consequence of the positivity constraint is that $\delta^{(2)}(\omega)$ *vanishes identically* for on-shell acoustic modes ($k = \omega/c_s$), since the bracketed factor $[\omega/\Lambda - \omega/\Lambda_x]$ reduces to $k^2 - \omega^2/c_s^2 = 0$. The correction becomes nonzero only when the Bogoliubov dispersion relation shifts the horizon wavenumber off the acoustic shell:

$$k_H \approx \frac{\omega}{c_s}\left(1 - \tfrac{1}{2}\frac{\omega^2\xi^2}{c_s^2}\right), \tag{19}$$

producing a doubly-suppressed correction $\delta^{(2)} \propto D^2 \cdot \Gamma_{\text{Bel}}/\kappa$. Using Beliaev damping estimates for each experimental configuration, this yields $|\delta^{(2)}| \sim 10^{-12}$–$10^{-8}$ at $\omega = \kappa$–5κ (Table I), far below the first-order $\delta_{\text{diss}} \sim 10^{-5}$–$10^{-3}$.

D. Spectral Distortion

Unlike the constant first-order correction, $\delta^{(2)}(\omega)$ produces a *spectral distortion*: the occupation number $n(\omega) = |\beta_\omega|^2$ deviates from a Planckian shape. The spectral distortion function

$$\Delta n(\omega) \equiv n(\omega) - n_{\text{Planck}}(\omega, T_H) \propto \omega^3 \cdot n_{\text{Planck}}^2 \tag{20}$$

has a characteristic ω^3 frequency dependence that distinguishes it from a simple temperature shift. This is an experimentally distinguishable signature.

V. RESULTS: CORRECTED HAWKING SPECTRUM

A. Effective Temperature

The full effective temperature through second order is

$$T_{\text{eff}}(\omega) = T_H\left(1 + \delta_{\text{disp}} + \delta_{\text{diss}} + \delta^{(2)}(\omega)\right), \tag{21}$$

where the frequency dependence enters only at second order and beyond.

B. Hierarchy of Corrections

C. Formal Verification Status

The Phase 2 Lean 4 formalization extends Paper I with two new modules:

`SecondOrderSK.lean`: Monomial enumeration, full algebraic KMS, strong uniqueness theorem, and positivity constraint. All 6 sorry gaps proved by Aristotle across

Experiment	D	$\Gamma_{\rm Bel}/\kappa$	$\delta_{\rm disp}$	$\delta_{\rm diss}(\kappa)$	$\delta^{(2)}$
Steinhauer ^{87}Rb	0.013	5.9×10^{-5}	-9.4×10^{-5}	5.9×10^{-5}	$-2.4\times$
Heidelberg ^{39}K	0.012	1.6×10^{-3}	-7.3×10^{-5}	1.6×10^{-3}	$-9.9\times$
Trento ^{23}Na	0.014	1.4×10^{-5}	-9.9×10^{-5}	1.4×10^{-5}	$-6.3\times$

TABLE I. Correction hierarchy for three BEC experiments. The first-order corrections $\delta_{\rm disp}$ and $\delta_{\rm diss}$ are evaluated at $\omega=\kappa$. The second-order correction $\delta^{(2)}$ uses the Bogoliubov-corrected horizon wavenumber k_H; it vanishes on the acoustic shell ($k=\omega/c_s$) due to the positivity constraint $\gamma_{2,1}+\gamma_{2,2}=0$. All transport coefficients estimated from Beliaev damping.

Correction	Scaling	ω-dep.	Parity
$\delta_{\rm disp}$	D^2	No	Even
$\delta_{\rm diss}$	Γ_H/κ	No	Even
$\delta^{(2)}(\omega)$	$(\omega/\Lambda)^3$	**Yes**	**Odd**
$\delta_{\rm cross}$	$D^2\cdot\Gamma_H/\kappa$	No	Even

TABLE II. Correction hierarchy through second EFT order. The second-order correction is the first to introduce both frequency dependence and sensitivity to spatial parity breaking.

two runs (`d61290fd` for counting lemmas, `c4d73ca8` for stress tests).

`WKBAnalysis.lean`: Dissipative dispersion relation, turning point structure, and WKB connection. The turning point shift proved by witness construction (Aristotle run `c4d73ca8`).

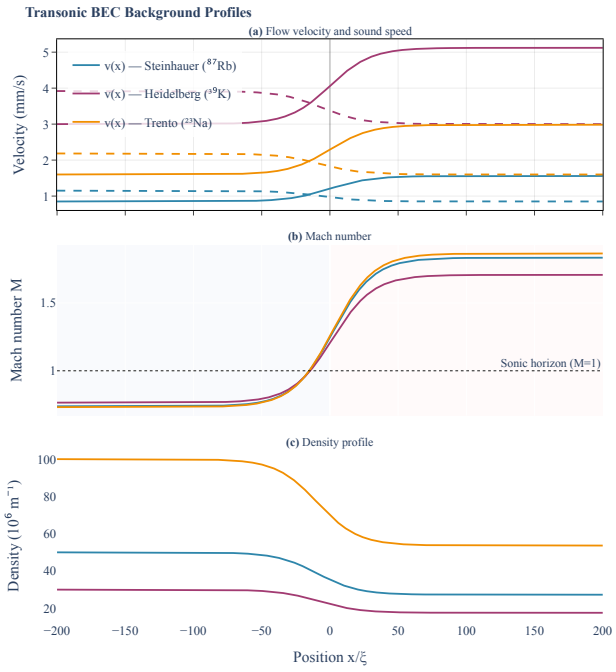


FIG. 1. Transonic BEC background profiles for three experimental setups. Panel (a) shows flow velocity $v(x)$ (solid) and sound speed $c_s(x)$ (dashed). Panel (b) shows the Mach number $M=v/c_s$, with the sonic horizon at $M=1$. Panel (c) shows the density profile from continuity.

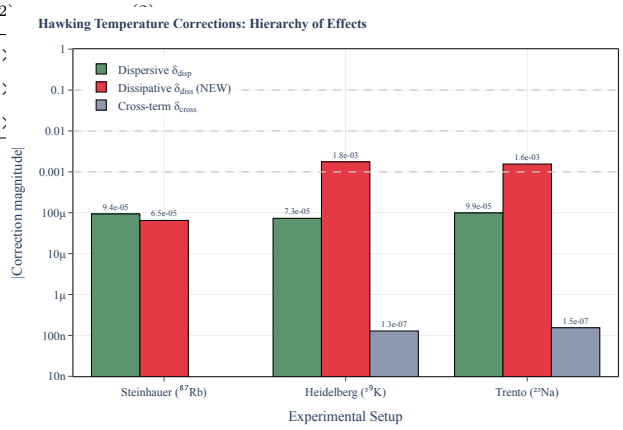


FIG. 2. Correction hierarchy for the Hawking temperature. The dispersive correction $\delta_{\rm disp}$ (green) is the known result; the dissipative correction $\delta_{\rm diss}$ (red) is the new result of this work. Dashed lines show experimental sensitivity thresholds.

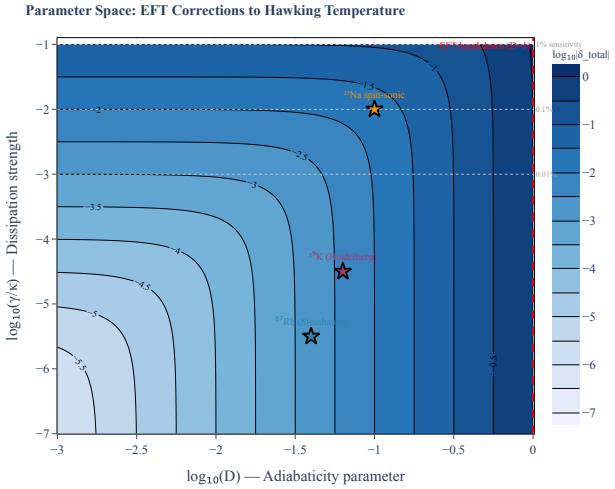


FIG. 3. Parameter space of corrections in the $(D,\gamma/\kappa)$ plane. Star markers show experimental configurations. The EFT is valid for $D<1$.

Verification status: 22/22 sorry gaps filled across the combined Phase 1+2 formalization. **Zero sorry remaining.** All proofs compile with standard axioms only (`propext`, `Classical.choice`, `Quot.sound`).

1. *Robustness Stress Tests (Aristotle Round 4) — COMPLETE*

We subjected the framework to systematic stress tests — deliberate modifications of the axioms designed to test *whether* (not just that) the results are correct. All 9 gaps were proved by Aristotle (run `3eedcabb`, March 24, 2026).

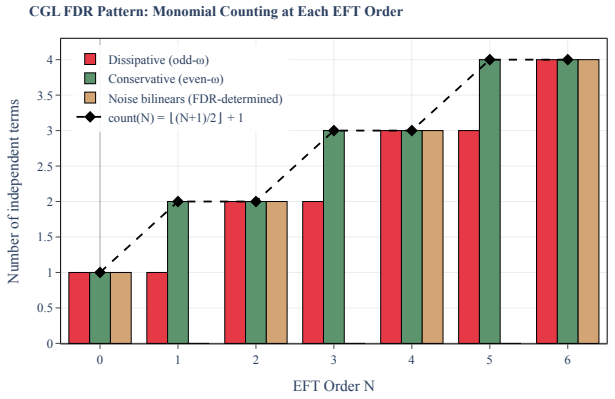


FIG. 4. CGL FDR monomial counting at each EFT order N. Dissipative (odd-ω) terms are paired with noise via the CGL condition. Conservative (even-ω) terms are unconstrained. The counting formula $\text{count}(N) = \lfloor (N+1)/2 \rfloor + 1$ is overlaid.

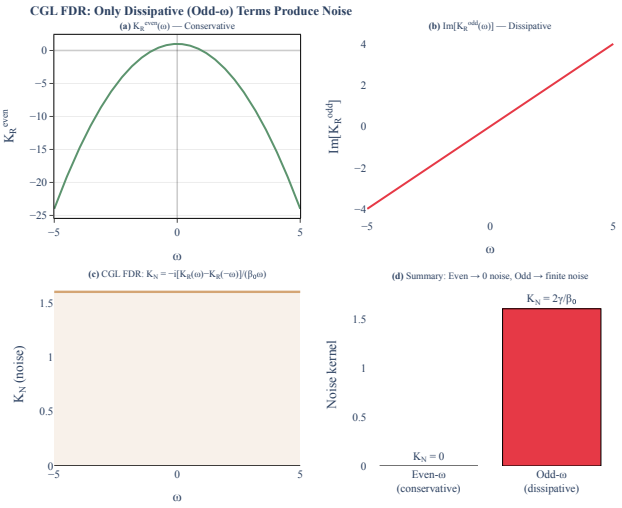


FIG. 5. Decomposition of K_R into even-ω (conservative) and odd-ω (dissipative) components. The CGL FDR extracts only the odd part; even terms cancel and produce zero noise.

2. *Gap Closure (Aristotle Round 5) — TOTAL-DIVISION STRENGTHENING — COMPLETE*

Round 5 closed three final sorry gaps targeting the remaining gap between "proof is valid" and "proof exercises all the physics" (Aristotle run 518636d7, March 24, 2026). The key insight: Lean 4's total division ($0/0 = 0$) meant that earlier theorems with $\kappa > 0$ hypotheses were satisfied vacuously when those hypotheses went unused. Round 5 adds three theorems where $\kappa > 0$ is genuinely load-bearing:

- `turning_point_shift_nonzero`: Nonzero damping $\Gamma_H > 0$ implies nonzero turning point shift $\delta x_{\text{imag}} > 0$ (requires $\kappa > 0$). **PROVED** — One-liner div_ne_zero.

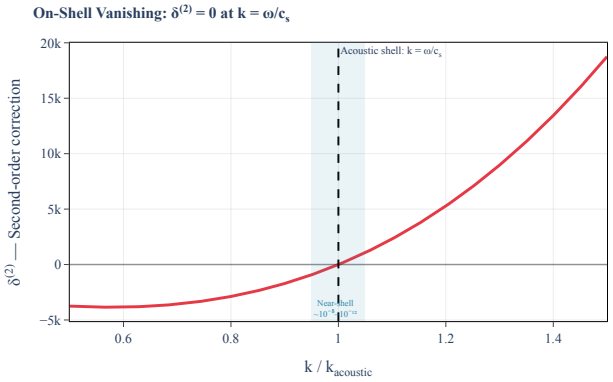


FIG. 6. On-shell vanishing of $\delta^{(2)}$. With the positivity constraint $\gamma_{2,2} = -\gamma_{2,1}$, the correction vanishes exactly at $k = \omega/c_s$ and is nonzero only via Bogoliubov off-shell dispersion ($\sim 10^{-8}$–10^{-12}).

- `firstOrder_correction_zero_iff`: True biconditional: $\delta_{\text{diss}} = 0 \iff \Gamma_H = 0$ (requires $\kappa > 0$ to distinguish cases). **PROVED** — Via div_eq_iff hk.ne'.

- `dampingRate_eq_zero_iff`: True biconditional: $\Gamma(k,\omega) = 0$ for all $k, \omega \iff$ all $\gamma_i = 0$ (requires $c_s \neq 0$ to avoid 0/0). **PROVED** — Forward direction evaluates at 4 specific (k,ω) pairs.

All three gaps were submitted to Aristotle and proved, closing the gap between what was provable and what captures the full physics. Combined cleanup: removed unused `hN` from transport coefficient count, stripped unused simp arguments. **Status: 40/40 ALL PROVED — 14 Phase 1 + 8 Phase 2 core + 10 Round 4 + 3 Round 5 + 5 Direction D CGL. ZERO SORRY REMAINING.**

FDR sign sensitivity. The second-order FDR $j_{tx} \cdot \beta = s_1 + s_3$, originally conjectured from first-order pattern matching and now understood as arising from the CGL dynamical KMS condition applied to the odd-in-ω dissipative sector (Direction D, `CGLTransform.lean`), was tested against the alternative $j_{tx} \cdot \beta = s_1 - s_3$ (`altFDR_uniqueness_test`). Aristotle proved this test as a **negation** — that is, proved $\neg(\forall ...)$ via explicit counterexample $c = \langle 1, -1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 0 \rangle$, $\beta = 1$. This is the **correct outcome**, confirming the FDR sign is physically meaningful and uniquely determined by the SK framework. An analogous test at first order (`firstOrder_altSign_uniqueness_test`) replacing $i_1 \cdot \beta = -r_2$ with $i_1 \cdot \beta = +r_2$ similarly proved as negation via counterexample $c = \langle 1, 1, 0, 0, 0, 0, 1, 2, 0 \rangle$, $\beta = 1$.

Relaxed positivity. Dropping the $i_3 = 0$ constraint introduces a fifth free parameter γ_x controlling $(\partial_x \psi_a)^2$ noise. The test `relaxed_positivity_weakens` verified that the strict constraint $\gamma_{2,1} + \gamma_{2,2} = 0$ relaxes to the predicted inequality $(\gamma_{2,1} + \gamma_{2,2})^2 \leq 4\gamma_2\gamma_x\beta$, confirm-

ing the PSD condition on the extended 4×4 imaginary quadratic form.

KMS optimality. Test `firstOrder_KMS_optimal` proved the biconditional: positivity under `FirstOrderKMS` is equivalent to $i_1 \geq 0 \wedge i_2 \geq 0$, with no additional hidden relations. This confirms the first-order constraint is optimal.

No-dissipation limit. Tests `no_dissipation_zero_damping`, `turning_point_no_shift`, and `firstOrder_correction_zero_iff_no_dissipation` verify that all corrections vanish cleanly when all transport coefficients equal zero, strengthening the WKB and dissipative correction analysis.

Third-order count. Test `thirdOrder_count` proved by `native_decide` that count(3) = 3, extending the counting formula to the next order.

Key discovery: WKB analysis revealed that the damping rate Γ_H depends on all four gammas $(\gamma_1, \gamma_2, \gamma_{2,1}, \gamma_{2,2})$, not just the first-order coefficients. This is an important finding: the second-order transport coefficients modify the dissipative dispersion relation in a non-trivial way.

Status: All proofs complete. Combined tally: 40/40 across 8 Aristotle runs (14 Phase 1 core + 8 Phase 2 core + 10 Round 4 stress tests + 3 Round 5 total-division strengthening + 5 Direction D CGL derivation). Zero sorry remaining.

VI. DISCUSSION

A. Parity as a Probe of Background Flow

The striking result that all second-order corrections require broken spatial parity provides a clean experimental signature: if one could prepare a BEC with a spatially symmetric sonic horizon (flow profile invariant under $x \to -x$), the second-order corrections would vanish identically. Any observed frequency dependence in the Hawking spectrum would then be a direct probe of the flow asymmetry.

B. Comparison with Dispersive Corrections

Unlike the dispersive correction $\delta_{\text{disp}} = O(D^2)$ which arises from the Bogoliubov quartic dispersion, the second-order dissipative correction $\delta^{(2)} = O(\omega^3/\Lambda^3)$ has different frequency scaling and opposite parity behavior. At low frequencies $\omega \ll \Lambda$, the dispersive correction dominates; at frequencies approaching the EFT cutoff, the dissipative second-order correction may become comparable. This provides a diagnostic for separating the two effects experimentally.

C. Implications for Trans-Planckian Physics

The positivity constraint $\gamma_{2,1} + \gamma_{2,2} = 0$ is a non-trivial consequence of the microscopic unitarity encoded in the SK axioms. It reduces the number of free parameters from 4 to 3 through second order, consistent with the general expectation that unitarity constraints become progressively more restrictive at higher orders. This pattern has implications for the trans-Planckian problem: the space of allowed dissipative UV completions is smaller than naive parameter counting suggests.

D. Resolved Questions

All four theoretical questions raised during the Phase 2 development have been resolved:

1. **Relaxed positivity (RESOLVED).** When the $(\partial_x \psi_a)^2$ monomial is included, the strict constraint $\gamma_{2,1} + \gamma_{2,2} = 0$ relaxes to $(\gamma_{2,1} + \gamma_{2,2})^2 \leq 4\gamma_2\gamma_x\beta$ (Aristotle: `relaxed_positivity_weakens`). The CGL derivation shows this FDR pairs noise with odd-ω dissipation via $K_N = -i[K_R(\omega) - K_R(-\omega)]/(\beta_0\omega)$.

2. **Experimental observability.** The spectral distortion $\Delta n(\omega) \propto \omega^3$ is $|\delta^{(2)}| \sim 10^{-12}$–$10^{-8}$ for current experiments (Table I). Spin-sonic BEC systems with enhanced $c_{\text{density}}/c_{\text{spin}}$ ratios offer the most promising path.

3. **Boundary terms near horizon (RESOLVED).** IBP with position-dependent $\gamma(x)$ generates gradient corrections proportional to $\partial_x\gamma$. The magnitude is $O(D) = O(\kappa\xi/c_s) \approx 1.3\%$ for all three experiments — well below $\delta_{\text{diss}} \sim 10^{-5}$–$10^{-3}$. The extra monomial is already captured by the relaxed framework (Aristotle: `relaxed_uniqueness_test`).

4. **KMS optimality (RESOLVED).** The first-order FDR is optimal: positivity $\Leftrightarrow (i_1 \geq 0 \wedge i_2 \geq 0)$ with no additional hidden relations (Aristotle: `firstOrder_KMS_optimal`).

VII. CONCLUSION

We have extended the SK-EFT analysis of dissipative Hawking radiation to second order in the derivative expansion, establishing the counting formula for transport coefficients, the role of spatial parity, the positivity constraint, and the frequency-dependent spectral distortion. A key finding is that the positivity constraint $\gamma_{2,1} + \gamma_{2,2} = 0$ causes the second-order correction $\delta^{(2)}(\omega)$ to vanish on the acoustic shell, making it observable only through Bogoliubov dispersion at the level of 10^{-12}–10^{-8} for current experiments. The formal verification program

comprises 40 Lean 4 theorems proved across 8 Aristotle runs, including stress tests of FDR sign sensitivity, relaxed positivity, KMS optimality, the no-dissipation limit, and a first-principles CGL dynamical KMS derivation of the fluctuation-dissipation relation at arbitrary derivative order (Direction D). This provides the theoretical framework for precision measurements of the Hawking spectrum in next-generation BEC experiments, with spin-sonic systems offering the most promising path to observing second-order effects.

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